

Resources:

- <https://www.whocanuse.com/>
- Web Content Accessibility Guidelines <https://www.w3.org/TR/WCAG21/>
- Vision Australia <https://www.visionaustralia.org/>
- <https://www.colour-blindness.com/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8069325/>
- <https://davidmathlogic.com/colorblind/#%23D81B60-%231E88E5-%23FFC107-%23004D40>

Color Conditions (ranked from most common to least common):

* highlighted = at least 1% of the population affected *

- Regular vision (trichromatic) [68% of worldwide population]
- Deuteranomaly (reduced sensitivity to green - trouble distinguishing reds and greens) [5.3%]
- Protanopia (red blind - can't see reds at all) [1.5%]
- Protanomaly (reduced sensitivity to red - trouble distinguishing reds and greens) [1.3%]
- Deuteranopia (green blind - can't see greens at all) [1.2%]
- Achromatomaly (partial color blindness, sees absence of most colors) [0.09%]
- Achromatopsia (complete color blindness, can only see shades) [0.05%]
- Tritanopia (unable to distinguish btw blues vs. greens, purples vs. reds, yellows vs. pinks) [0.03%]
- Tritanomaly (trouble distinguishing blues vs. greens, yellows vs. reds) [0.02%]

Other Vision Conditions:

- Cataracts (clouding of eye lens) [33%]
- Low vision (decreased/blurry vision not fixable by usual means such as glasses) [31%]
- Glaucoma (slight vision loss) [2%]
- Direct sunlight on phone/screen
- Night shift mode

<https://www.colour-blindness.com/> notes

- red green color blind (most common - 95% of colorblind people)
 - dichromacy (only two color receptors) - protanopia and deuteranopia
 - anomalous trichromacy (slight defect, where proportions of color receptors are different) - protanomaly and deuteranomaly
- blue yellow color blind (5% of colorblind people)
 - dichromacy - tritanopia
 - anomalous trichromacy - tritanomaly
- total colorblindness (least common)
- green deficiency - mildest type, most common type of colorblindness
- total inability to see red - second most common

Thoughts on Algorithms?

- Start with ROYGBIV mappings to acceptable colors (that work for all the conditions that more than 1% of population has) → need to be more oriented around pairs/combinations of colors
- <https://github.com/skrachdot/color-blind> takes in a hexcode and returns RGB values (or hexcode) of what the colorblind person with the condition sees

- Check the effectiveness of our iterations by passing our color-corrected image here
- detect top 4-5 dominant colors in the input image
- if colors are similar by a certain threshold, switch out one of the colors (get closest ROYGBIV color to the dominant color and use that as a starting point)
 - can decide what to switch the color to by:
 - table of distinct colors
- Starting idea: bucket all the colors into rough RGB ranges → if they're in different buckets in the actual image but the same bucket in the converted image, they need to be assigned different colors

Techniques for Image Transformation

- Goal: Preservation of color naturalness + Preservation/Enhancement of contrast
- Fuzzy clustering-based color segmentation to extract key colors
- Mapping of key colors to CIE 1931 chromaticity diagram
- Translate key colors on the same confusion line to different confusion lines
- Objective function that can be optimized to drive adaptation of key colors
- Rotate color plane
- Maintain saturation and brightness while applying nonlinear mapping to hue
 - Specifically, the ranges of reds and greens are translated towards the ranges of blues and yellows, which are distinguishable by the protanopes and deuteranopes